8. AdHoc QA Agent

Overview

The Ad Hoc Q&A Agent is a conversational agent used after a Unit Risk Assessment is generated in the workflow. It allows Reliability Engineers (RE) and Outage Engineers (OE) to ask follow-up questions and clarify findings from the assessment. The agent answers questions by reasoning over risk assessment outputs (risk assessment results, narrative summaries, event history) and underlying source data (e.g., IBAT, PRISM, FSR/ER documents). The goal is to help engineers quickly understand why a risk was flagged, explore supporting evidence, and investigate related operational history.

Who Uses It:

Reliability Engineers (REL) - investigate long-term reliability trends, FMEA findings, and inspection history. The agent surfaces RE-specific risk tables, reports, and inspection data.
Outage Engineers (OE) - focus on outage planning, event history, and operational readiness. The agent surfaces OE-specific tables, event master records, and event reports.

What it reasons over

- Risk Assessment output for the unit
- Narrative Summary generated by the prior workflow step
- Inspection and operating history (IBAT)
- FMEA / failure mode data (PRISM)
- Field Service Reports and Engineering Requests (FSR, ER)
- Risk matrix, issue data

Key Behaviors

- Session key = Assessment ID, but chat history is stored within a user-scoped namespace. In practice, sessions are assessment-scoped within a user scope, not globally shared across all users for the same assessment.
- Answers are grounded in tool results - the agent does not hallucinate or speculate beyond what the data shows.
- Persona-aware - the tool set available is determined by the engineer's role (RE or OE)

Where it sits in the workflow

User Submits ESN

Displays doc list (FSR/ER...)

User uploads missing docs

Runs Risk Assessment

Risk Assessment Issues Table

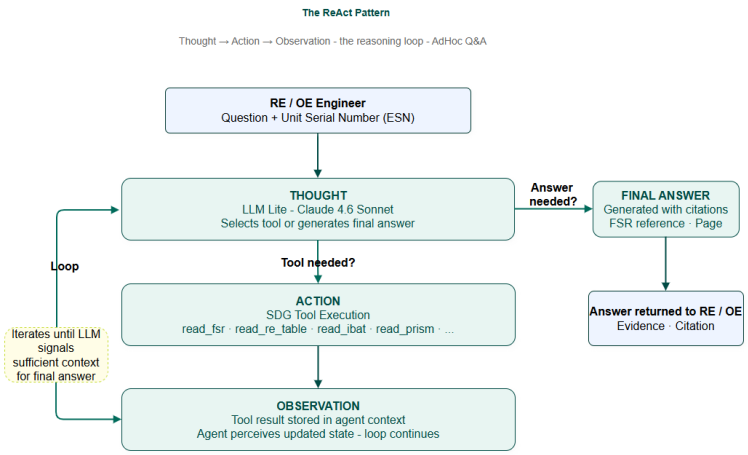
Feedback (Human in the loop)

Narrative Summary

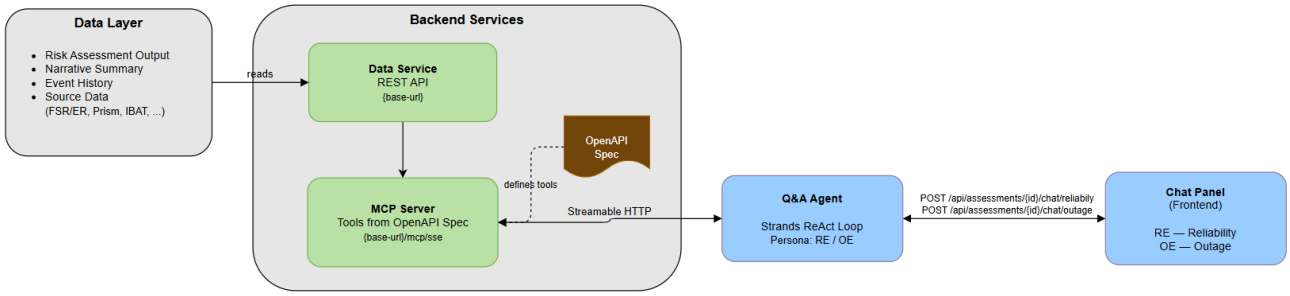
Ad Hoc Q&A Chat (RE / OE) this agent

How the Agent Reasons - ReAct Loop

The Agent uses Strands framework which manages the ReAct loop (Thought Action Observe).



Architecture



Agent Architecture Diagram.drawio

Technology Stack

Component	Choice
Agentic Framework	Strands (native agentic loop)
Workflow	Strands
LLM	Claude 4.6 Sonnet
API	Fast API, Streamable HTTP
Session Memory	`S3SessionManager` (Strands internal)
Tools	Data services exposed via MCP
Access Control	Persona-based tool list on agent construction
Observability	CloudWatch Logs
Container	ECS Fargate

Persona Based Tool Access

Tool	RE	OE	Underlying Operation ID	Data Services API Mapping

read_ibat			get_ibat_equipment_endpoint	GET /dataservices/api/v1/ibat/equipment
read_prism		—	read_prism	POST /dataservices/api/v1/prism/read
query_fsr			retrieve_endpoint	POST /dataservices/api/v1/retriever/retrieve
query_er			get_risk_er_cases_endpoint	POST /dataservices/api/v1/er/risk-er-cases
read_risk_matrix			load_heatmap	GET /dataservices/api/v1/heatmap/load
read_assessment_table			get_findings	GET /dataservices/api/v1/assessments/{assessment_id}/findings
read_re_narrative			get_re_narrative	GET /dataservices/api/v1/assessments/{assessment_id}/re-narrative
read_event_master	—		get_event_master	GET /dataservices/api/v1/events/master
read_event_history	—		get_event_history	GET /dataservices/api/v1/assessments/{assessment_id}/event-history

System Prompts

	Files shared by DS team	**Git Link**
RE	system_prompt_adhoc_agent_RE.txt	RE System Prompt Git Link
OE	system_prompt_adhoc_agent_OE.txt	OE System Prompt Git Link